

Calorimeter pointing: v4 Stage 1 preparation

Stored inputs, offline smearing and the bounded signal check

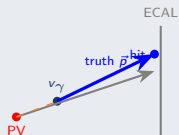
FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ study

5 October 2026

PI checkpoint: prepare reproducible Zbb, Zcc, Zss and 100k forced-mode reprocessing.

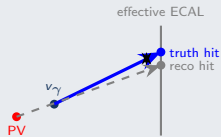
Pointing study: four steps

1. Save the photon



Keep E_{reco} , $\vec{p}_{reco}$, matched truth $\vec{p}_\gamma$ and production point. Reco direction currently assumes the photon starts at the origin.

2. Predict both hits



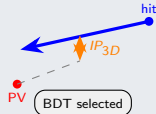
Intersect the truth ray from v_γ with the cylinder/endcap. Also save the origin-based reco projection as a position-response comparison.

3. Smear the direction



Add two Gaussian angular components in the tangent plane. Scan $\sigma = 0.5, 0.7, 0.9, 1.1, 1.3, 1.5$ mrad. Same photon, same random draw across candidates.

4. Measure after BDT



Set $\vec{p}^{hyp} = E_{reco} \hat{n}_{smear}$. Compute IP_{3D} and $|d_0|$ to the reconstructed PV; scan cuts and signal/background efficiencies offline.

The truth-hit picture assumes an ideal hit position; the reco-hit picture inherits current position uncertainty. Neither assigns a pointing significance without a covariance model.

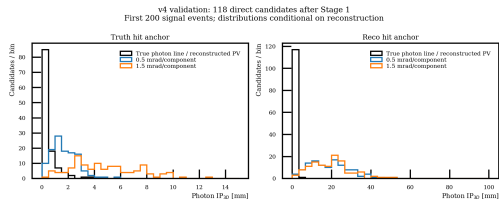
What is stored and how pointing hypotheses are made

For each reconstructed Λ_b candidate the v4 tuple keeps:

- measured photon $E_{\text{reco}}, (p_x, p_y, p_z)_{\text{reco}}$ and reco index;
- matched truth photon (p_x, p_y, p_z, E) and production vertex;
- predicted ECAL hit from the displaced truth ray and, separately, from the current origin-based reco direction;
- reconstructed PV and effective surface parameters.

Smear the truth direction in two orthogonal tangent directions with Gaussian component width $\sigma = 0.5\text{--}1.5$ mrad (six points), then set $\vec{p}_{\gamma}^{\text{hyp}} = E_{\text{reco}} \hat{n}_{\text{smear}}$. Reuse the same random draw for a photon shared by candidates. Evaluate $IP_{3D} = |(\vec{x}_{\text{hit}} - \vec{x}_{PV}) \times \hat{n}|$ and $|d_0|$ offline after BDT.

Effective IDEA geometry: barrel $R = 2250$ mm, $|z| = 2500$ mm; endcap inner radius inferred as $2500 / \sinh(3)$. A pointing significance needs a position/direction covariance not yet modelled.



118 matched direct candidates from the first 200 signal input events. Conditional shapes, not yields. Truth-hit anchor assumes ideal position; reco anchor inherits current position uncertainty.

What the 200-event signal check established

Input events	200	Candidate-bearing events	121
Candidates	133	Direct / wrong combinations	118 / 15
Old branches identical	474 / 474	Pointing vectors valid	28 / 28
Stage 2 audit / selected	133 / 129	After frozen BDT cut	46

What passed

The true-ray projection closes to its saved hit. The new fields survive Stage 2 preparation and the frozen BDT score. The paired run kept the same 121 output events and the same 133 candidates.

What remains to measure

Full signal and Zss/Zcc processing must establish unresolved fractions, pointing-cut efficiencies, mass and angle sculpting, and signal acceptance. This small signal check does not measure flavour rejection.

v4 Condor check only

Zbb: 4,398 files; Zcc: 5,018; Zss: 5,015. Each has 1,200 chunks, four cores/job, accounting group `group_u_FCC.local_gen` and `workday` flavour. All three checks passed; no jobs were submitted.

100k Stage 1 rerun: available modes and next commands

Mode	Current merged input	Status
$\Lambda\gamma$ PHSP	Lb2LambdaGamma_nev100000...root	ready
$\Lambda\gamma$ HELAMP	Lb2LambdaGammaPhysics_nev100000...root	ready
$\Lambda\eta$ PHSP / HELAMP	Lb2LambdaEta(Physics)_nev100000...root	ready
$\Lambda\pi^0$ HELAMP	Lb2LambdaPi0Physics_nev100000...root	ready
$\Lambda\pi^0$ PHSP	Lb2LambdaPi0_nev100000...root	merged file not found

- 1 Run `scripts/run_stage1_v4_forced_100k.sh --check-only`; it lists exact inputs/outputs. Then use `--run` to process available modes. Default uses four concurrent samples $\times$ eight CPUs (32 cores); tune `V4_DIRECT_JOBS` and `V4_DIRECT_CPUS`. Each output gets a v4 tuple validation JSON.
- 2 Generate/merge the missing π^0 PHSP 100k sample using seed base 72001 from `config/config.yaml`, then rerun the checker.
- 3 After review, submit Z flavours with `scripts/submit_stage1_v4_zflavours.sh --submit`; default is `--check-only`.

Production commands, output paths, file inventory and check logs are recorded in the v4 howto and frozen job preparation note. No full 100k or Condor production was launched here.